



MHH772 / 90DaysOfDevOps



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90DaysOfDevOps / 2026 / day-20



LondheShubham153 Added day 20 challenge

8fde409 · 7 months ago



This branch is 17 commits ahead of TrainWithShubham/90DaysOfDevOps:master



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Sync fork

Name	Name	Last commit date
..		
README.md	Added day 20 challenge	7 months ago
sample_logs_generato...	Added day 20 challenge	7 months ago

README.md



Day 20 – Bash Scripting Challenge: Log Analyzer and Report Generator

Task

You are a system administrator responsible for managing a network of servers. Every day, a log file is generated on each server containing important system events and error messages. Your job is to analyze these log files, identify specific events, and generate a summary report.

Write a Bash script (`log_analyzer.sh`) that automates the process of analyzing log files and generating a daily summary report.

Expected Output

- A Bash script: `log_analyzer.sh`
- A generated summary report: `log_report_<date>.txt`
- A markdown file: `day-20-solution.md` documenting your approach

Challenge Tasks

Task 1: Input and Validation

Your script should:

1. Accept the path to a log file as a command-line argument
2. Exit with a clear error message if no argument is provided
3. Exit with a clear error message if the file doesn't exist

Task 2: Error Count

1. Count the total number of lines containing the keyword `ERROR` or `Failed`
2. Print the total error count to the console

Task 3: Critical Events

2. Print those lines along with their line number

Example output:

```
--- Critical Events ---  
Line 84: 2025-07-29 10:15:23 CRITICAL Disk space below threshold  
Line 217: 2025-07-29 14:32:01 CRITICAL Database connection lost
```



Task 4: Top Error Messages

1. Extract all lines containing `ERROR`
2. Identify the **top 5 most common** error messages
3. Display them with their occurrence count, sorted in descending order

Example output:

```
--- Top 5 Error Messages ---
45 Connection timed out
32 File not found
28 Permission denied
15 Disk I/O error
9 Out of memory
```



Task 5: Summary Report

Generate a summary report to a text file named `log_report_<date>.txt` (e.g., `log_report_2026-02-11.txt`). The report should include:

1. Date of analysis
2. Log file name
3. Total lines processed
4. Total error count
5. Top 5 error messages with their occurrence count
6. List of critical events with line numbers

Task 6 (Optional): Archive Processed Logs

Add a feature to:

1. Create an `archive/` directory if it doesn't exist
2. Move the processed log file into `archive/` after analysis
3. Print a confirmation message

Sample Log File

A sample log file is available in this directory: `sample_log.log`

You can also pick real-world log datasets from the [LogHub repository](#) to test your script against production-like logs (e.g., ZooKeeper, HDFS, Apache, Linux syslogs).

Hints

- Count errors: `grep -c "ERROR" logfile.log`
- Print with line numbers: `grep -n "CRITICAL" logfile.log`
- Top occurrences: `grep "ERROR" logfile.log | awk '{$1=$2=$3=""; print}' | sort | uniq -c | sort -rn | head -5`
- Associative arrays: `declare -A error_map`
- Date for filename: `date +%Y-%m-%d`
- Move files: `mv logfile.log archive/`

Documentation

Create `day-20-solution.md` with:

- Your script's code
- Sample output from running against the sample log
- What commands/tools you used (`grep` , `awk` , `sort` , `uniq` , etc.)
- What you learned (3 key points)

Submission

1. Add your scripts and `day-20-solution.md` to `2026/day-20/`
2. Commit and push to your fork

Learn in Public

Share your log analyzer project on LinkedIn.

`#90DaysOfDevOps` `#DevOpsKaJosh` `#TrainWithShubham`

Happy Learning! **TrainWithShubham**